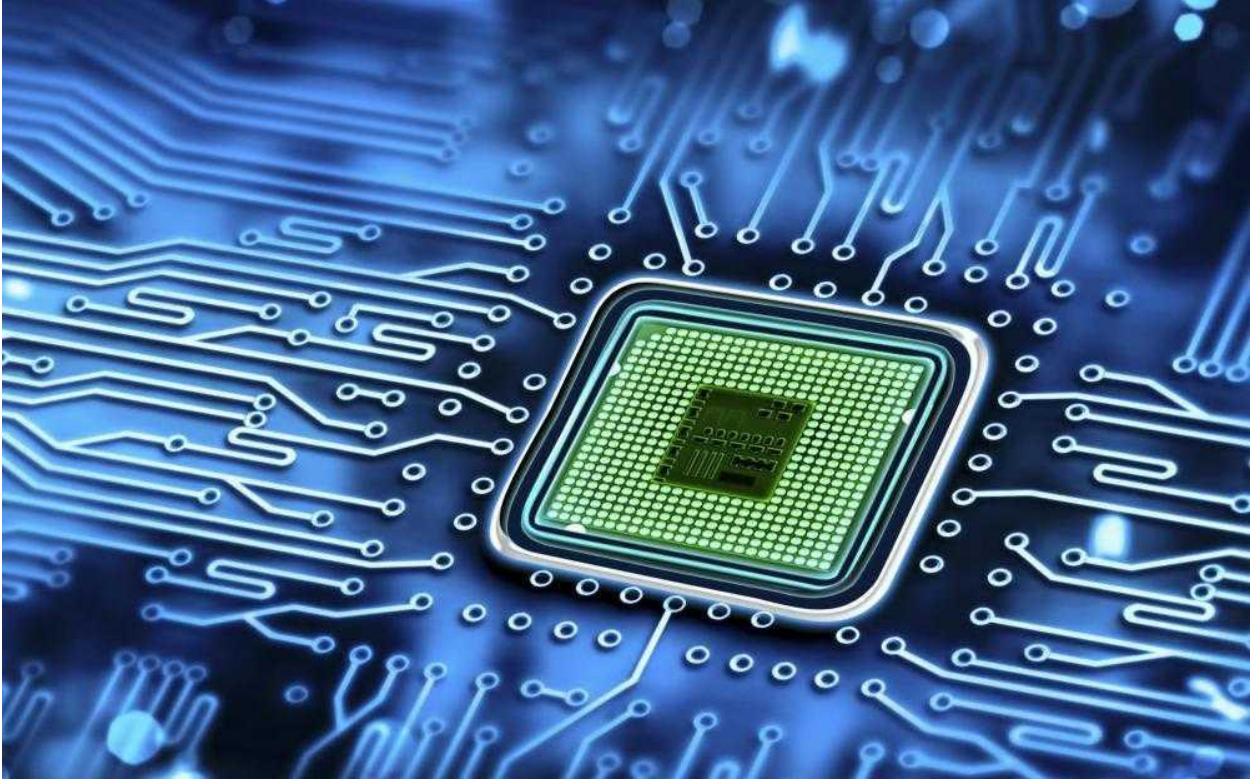


Project1

DSP48A1



Name: Arwa Ashraf Kantoush

Q1)

Verilog Code

RTL Code

DSP_REG

```
module DSP_REG #(
    parameter WIDTH = 18,
    parameter REG = 1,
    parameter RSTTYPE = "SYNC"
) (
    input CLK,
    input rst,
    input ce,
    input [WIDTH-1:0]in,
    output reg [WIDTH-1:0]out
);

generate
    if (REG && RSTTYPE == "SYNC") begin : GEN_SYNC
        always @(posedge CLK) begin
            if (rst) begin
                out <= 0;
            end
            else if (ce) begin
                out <= in;
            end
        end
    end
    else if (REG && RSTTYPE == "ASYNC") begin : GEN_ASYNC
        always @(posedge CLK or posedge rst) begin
            if (rst) begin
                out <= 0;
            end
            else if (ce) begin
                out <= in;
            end
        end
    end
    else begin : GEN_BYPASS
        always @(*) begin
            out = in;
        end
    end
end
```

```

        end
    end
endgenerate

```

```
endmodule //DSP_reg
```

DSP

```

module DSP #(
    parameter WIDTH_18 = 18,
    parameter WIDTH_48 = 48,
    parameter A0REG = 0,
    parameter A1REG = 1,
    parameter B0REG = 0,
    parameter B1REG = 1,
    parameter CREG = 1,
    parameter DREG = 1,
    parameter MREG = 1,
    parameter PREG = 1,
    parameter CARRYINREG = 1,
    parameter CARRYOUTREG = 1,
    parameter OPMODEREG = 1,
    parameter CARRYINSEL = "OPMODE5",
    parameter B_INPUT = "DIRECT",
    parameter RSTTYPE = "SYNC"
) (
    input CLK,
    input [7:0]OPMODE,
    input CEA,
    input CEB,
    input CEC,
    input CECARRYIN,
    input CED,
    input CEM,
    input CEOPMODE,
    input CEP,
    input RSTA,
    input RSTB,
    input RSTC,
    input RSTCARRYIN,
    input RSTD,
    input RSTM,
    input RSTOPMODE,
    input RSTP,
    input [WIDTH_18-1:0]A,
    input [WIDTH_18-1:0]B,

```

```

    input [WIDTH_18-1:0]BCIN,
    input [WIDTH_48-1:0]C,
    input [WIDTH_18-1:0]D,
    input CARRYIN,
    input [WIDTH_48-1:0]PCIN,
    output [35:0]M,
    output [WIDTH_48-1:0]P,
    output CARRYOUT,
    output CARRYOUTF,
    output [WIDTH_18-1:0]BCOUT,
    output [WIDTH_48-1:0]PCOUT
);

wire [WIDTH_18-1:0]A0_REG,A1_REG,B_SEL,B0_REG,B1_REG,D_REG,SUM1,SUM1_SEL;
wire [35:0]MUL,M_REG;
wire [WIDTH_48-1:0]C_REG;
wire [WIDTH_48:0]P_REG;
reg [WIDTH_48-1:0]SEL_X,SEL_Z;
wire CYI_SEL,CYI;
wire [7:0]OPMODE_REG;

DSP_REG #(.WIDTH(8),.REG(OPMODEREG),.RSTTYPE(RSTTYPE)) dutOPMODE
(.CLK(CLK),.rst(RSTOPMODE),.ce(CEOPMODE),.in(OPMODE),.out(OPMODE_REG));
DSP_REG #(.WIDTH(WIDTH_18),.REG(DREG),.RSTTYPE(RSTTYPE)) dutD
(.CLK(CLK),.rst(RSTD),.ce(CED),.in(D),.out(D_REG));

generate
    if (B_INPUT == "DIRECT") begin : GEN_B_DIRECT
        assign B_SEL = B;
    end
    else if (B_INPUT == "CASCADE") begin : GEN_B_CASCADE
        assign B_SEL = BCIN;
    end
    else begin : GEN_B_ZERO
        assign B_SEL = 18'b0;
    end
endgenerate

DSP_REG #(.WIDTH(WIDTH_18),.REG(B0REG),.RSTTYPE(RSTTYPE)) dutB0
(.CLK(CLK),.rst(RSTB),.ce(CEB),.in(B_SEL),.out(B0_REG));
assign SUM1 = (OPMODE_REG[6] == 1'b0)? (D_REG + B0_REG) : (D_REG - B0_REG);
assign SUM1_SEL = (OPMODE_REG[4] == 1'b0)? B0_REG : SUM1;
DSP_REG #(.WIDTH(WIDTH_18),.REG(B1REG),.RSTTYPE(RSTTYPE)) dutB1
(.CLK(CLK),.rst(RSTB),.ce(CEB),.in(SUM1_SEL),.out(B1_REG));

```

```

DSP_REG #(.WIDTH(WIDTH_18),.REG(A0REG),.RSTTYPE(RSTTYPE)) dutA0
(.CLK(CLK),.rst(RSTA),.ce(CEA),.in(A),.out(A0_REG));
DSP_REG #(.WIDTH(WIDTH_18),.REG(A1REG),.RSTTYPE(RSTTYPE)) dutA1
(.CLK(CLK),.rst(RSTA),.ce(CEA),.in(A0_REG),.out(A1_REG));
assign MUL = B1_REG * A1_REG;
DSP_REG #(.WIDTH(36),.REG(MREG),.RSTTYPE(RSTTYPE)) dutM
(.CLK(CLK),.rst(RSTM),.ce(CEM),.in(MUL),.out(M_REG));
always @(*) begin
    case (OPMODE_REG[1:0])
        2'b00: SEL_X = 0;
        2'b01: SEL_X = {12'b0,M_REG};
        2'b10: SEL_X = P;
        2'b11: SEL_X = {D_REG[11:0],A1_REG[17:0],B1_REG[17:0]};
        default : SEL_X = 48'b0;
    endcase
end
DSP_REG #(.WIDTH(WIDTH_48),.REG(CREG),.RSTTYPE(RSTTYPE)) dutC
(.CLK(CLK),.rst(RSTC),.ce(CEC),.in(C),.out(C_REG));
always @(*) begin
    case (OPMODE_REG[3:2])
        2'b00: SEL_Z = 0;
        2'b01: SEL_Z = PCIN;
        2'b10: SEL_Z = P;
        2'b11: SEL_Z = C_REG;
        default : SEL_Z = 48'b0;
    endcase
end
generate
    if (CARRYINSEL == "OPMODE5") begin : GEN_CYI_OPMODE5
        assign CYI_SEL = OPMODE_REG[5];
    end
    else if (CARRYINSEL == "CARRYIN") begin : GEN_CYI_CARRYIN
        assign CYI_SEL = CARRYIN;
    end
    else begin : GEN_CYI_ZERO
        assign CYI_SEL = 1'b0;
    end
endgenerate
DSP_REG #(.WIDTH(1),.REG(CARRYINREG),.RSTTYPE(RSTTYPE)) dutCYI
(.CLK(CLK),.rst(RSTCARRYIN),.ce(CECARRYIN),.in(CYI_SEL),.out(CYI));
assign P_REG = (OPMODE_REG[7] == 1'b0)? (SEL_Z + SEL_X + CYI) : (SEL_Z - (SEL_X +
CYI));
DSP_REG #(.WIDTH(WIDTH_48),.REG(PREG),.RSTTYPE(RSTTYPE)) dutP
(.CLK(CLK),.rst(RSTP),.ce(CEP),.in(P_REG[WIDTH_48-1:0]),.out(P));

```

```

DSP_REG #(.WIDTH(1),.REG(CARRYOUTREG),.RSTTYPE(RSTTYPE)) dutCYO
(.CLK(CLK),.rst(RSTCARRYIN),.ce(CECARRYIN),.in(P_REG[WIDTH_48]),.out(CARRYOUT));
assign CARRYOUTF = CARRYOUT;
assign M = M_REG;
assign BCOUT = B1_REG;
assign PCOUT = P;

endmodule //DSP

```

Testbench code

```

module DSP_tb ();
parameter WIDTH_18 = 18;
parameter WIDTH_48 = 48;
parameter A0REG = 0;
parameter A1REG = 1;
parameter B0REG = 0;
parameter B1REG = 1;
parameter CREG = 1;
parameter DREG = 1;
parameter MREG = 1;
parameter PREG = 1;
parameter CARRYINREG = 1;
parameter CARRYOUTREG = 1;
parameter OPMODEREG = 1;
parameter CARRYINSEL = "OPMODE5";
parameter B_INPUT = "DIRECT";
parameter RSTTYPE = "SYNC";

reg CLK;
reg [7:0]OPMODE;
reg CEA;
reg CEB;
reg CEC;
reg CECARRYIN;
reg CED;
reg CEM;
reg CEOPMODE;
reg CEP;
reg RSTA;
reg RSTB;
reg RSTC;
reg RSTCARRYIN;
reg RSTD;
reg RSTM;
reg RSTOPMODE;

```

```

reg RSTP;
reg [WIDTH_18-1:0]A;
reg [WIDTH_18-1:0]B;
reg [WIDTH_18-1:0]BCIN;
reg [WIDTH_48-1:0]C;
reg [WIDTH_18-1:0]D;
reg CARRYIN;
reg [WIDTH_48-1:0]PCIN;
wire [35:0]M;
wire [WIDTH_48-1:0]P;
wire CARRYOUT;
wire CARRYOUTF;
wire [WIDTH_18-1:0]BCOUT;
wire [WIDTH_48-1:0]PCOUT;
reg [35:0]M_exp;
reg [WIDTH_48-1:0]P_exp;
reg CARRYOUT_exp;
reg CARRYOUTF_exp;
reg [WIDTH_18-1:0]BCOUT_exp;
reg [WIDTH_48-1:0]PCOUT_exp;

DSP #(
    .WIDTH_18(WIDTH_18),
    .WIDTH_48(WIDTH_48),
    .A0REG(A0REG),
    .A1REG(A1REG),
    .B0REG(B0REG),
    .B1REG(B1REG),
    .CREG(CREG),
    .DREG(DREG),
    .MREG(MREG),
    .PREG(PREG),
    .CARRYINREG(CARRYINREG),
    .CARRYOUTREG(CARRYOUTREG),
    .OPMODEREG(OPMODEREG),
    .CARRYINSEL(CARRYINSEL),
    .B_INPUT(B_INPUT),
    .RSTTYPE(RSTTYPE)
) dut (
    .CLK(CLK),
    .OPMODE(OPMODE),
    .CEA(CEA),
    .CEB(CEB),
    .CEC(CEC),
    .CECARRYIN(CECARRYIN),

```

```

        .CED(CED),
        .CEM(CEM),
        .CEOPMODE(CEOPMODE),
        .CEP(CEP),
        .RSTA(RSTA),
        .RSTB(RSTB),
        .RSTC(RSTC),
        .RSTCARRYIN(RSTCARRYIN),
        .RSTD(RSTD),
        .RSTM(RSTM),
        .RSTOPMODE(RSTOPMODE),
        .RSTP(RSTP),
        .A(A),
        .B(B),
        .BCIN(BCIN),
        .C(C),
        .D(D),
        .CARRYIN(CARRYIN),
        .PCIN(PCIN),
        .M(M),
        .P(P),
        .CARRYOUT(CARRYOUT),
        .CARRYOUTF(CARRYOUTF),
        .BCOUT(BCOUT),
        .PCOUT(PCOUT)
    );

initial begin
    CLK = 1;
    forever begin
        #1 CLK = ~CLK;
    end
end

initial begin
    //1
    RSTA = 1; RSTB = 1; RSTC = 1; RSTCARRYIN = 1; RSTD = 1; RSTM = 1; RSTOPMODE =
1; RSTP = 1;
    CEA = $random; CEB = $random; CEC = $random; CECARRYIN = $random; CED =
$random; CEM = $random; CEOPMODE = $random; CEP = $random;
    A = $random; B = $random; BCIN = $random; C = $random; D = $random; OPMODE =
$random; CARRYIN = $random; PCIN = $random;
    M_exp = 0; P_exp = 0; CARRYOUT_exp = 0; CARRYOUTF_exp = 0; BCOUT_exp = 0;
PCOUT_exp = 0;
    @(negedge CLK);

```



```

    if (M!==(M_exp || P!==(P_exp || CARRYOUT!==(CARRYOUT_exp ||
CARRYOUTF!==(CARRYOUTF_exp || BCOUT!==(BCOUT_exp || PCOUT!==(PCOUT_exp) begin
        $display("ERROR!");
        $stop;
    end
    //2
    RSTA = 0; RSTB = 0; RSTC = 0; RSTCARRYIN = 0; RSTD = 0; RSTM = 0; RSTOPMODE =
0; RSTP = 0;
    CEA = 1; CEB = 1; CEC = 1; CECARRYIN = 1; CED = 1; CEM = 1; CEOPMODE = 1; CEP
= 1;
    A = 20; B = 10; BCIN = $random; C = 350; D = 25; OPMODE = 8'b11011101;
CARRYIN = $random; PCIN = $random;
    M_exp = 'h12c; P_exp = 'h32; CARRYOUT_exp = 0; CARRYOUTF_exp = 0; BCOUT_exp =
'hf; PCOUT_exp = 'h32;
    repeat(4) @(negedge CLK);
    if (M!==(M_exp || P!==(P_exp || CARRYOUT!==(CARRYOUT_exp ||
CARRYOUTF!==(CARRYOUTF_exp || BCOUT!==(BCOUT_exp || PCOUT!==(PCOUT_exp) begin
        $display("ERROR!");
        $stop;
    end
    //3
    OPMODE = 8'b00010000;
    M_exp = 'h2bc; P_exp = 'h0; CARRYOUT_exp = 0; CARRYOUTF_exp = 0; BCOUT_exp =
'h23; PCOUT_exp = 'h0;
    repeat(3) @(negedge CLK);
    if (M!==(M_exp || P!==(P_exp || CARRYOUT!==(CARRYOUT_exp ||
CARRYOUTF!==(CARRYOUTF_exp || BCOUT!==(BCOUT_exp || PCOUT!==(PCOUT_exp) begin
        $display("ERROR!");
        $stop;
    end
    //4
    OPMODE = 8'b00001010;
    M_exp = 'hc8; P_exp = 'h0; CARRYOUT_exp = 0; CARRYOUTF_exp = 0; BCOUT_exp =
'ha; PCOUT_exp = 'h0;
    repeat(3) @(negedge CLK);
    if (M!==(M_exp || P!==(P_exp || CARRYOUT!==(CARRYOUT_exp ||
CARRYOUTF!==(CARRYOUTF_exp || BCOUT!==(BCOUT_exp || PCOUT!==(PCOUT_exp) begin
        $display("ERROR!");
        $stop;
    end
    //5
    A = 5; B = 6; C = 350; D = 25; PCIN = 3000;
    OPMODE = 8'b10100111;
    M_exp = 'h1e; P_exp = 'hfe6fffec0bb1; CARRYOUT_exp = 1; CARRYOUTF_exp = 1;
BCOUT_exp = 'h6; PCOUT_exp = 'hfe6fffec0bb1;

```

```

        repeat(3) @(negedge CLK);
        if (M!=M_exp || P!=P_exp || CARRYOUT!=CARRYOUT_exp ||
CARRYOUTF!=CARRYOUTF_exp || BCOUT!=BCOUT_exp || PCOUT!=PCOUT_exp) begin
            $display("ERROR!");
            $stop;
        end
        $stop;
end

initial begin
    $monitor("M=%h,M_exp=%h,P=%h,P_exp=%h,PCOUT=%h,PCOUT_exp=%h,CARRYOUT=%h,CARRY
OUT_exp=%h,CARRYOUTF=%h,CARRYOUTF_exp=%h,BCOUT=%h,BCOUT_exp=%h",M,M_exp,P,P_exp,P
COUT,PCOUT_exp,CARRYOUT,CARRYOUT_exp,CARRYOUTF,CARRYOUTF_exp,BCOUT,BCOUT_exp);
end
endmodule //DSP_tb

```

Simulation Tool

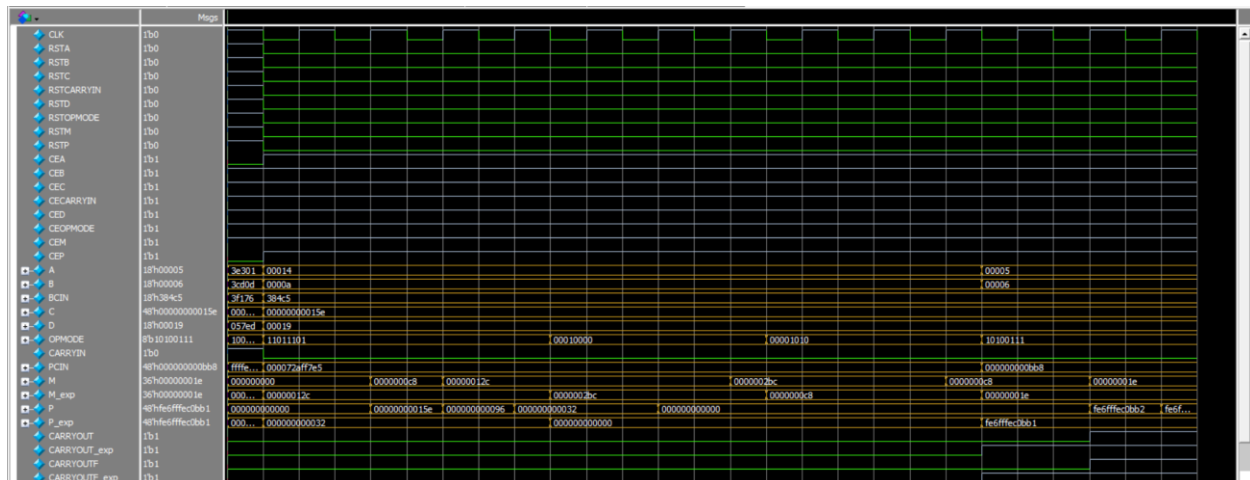
Do file

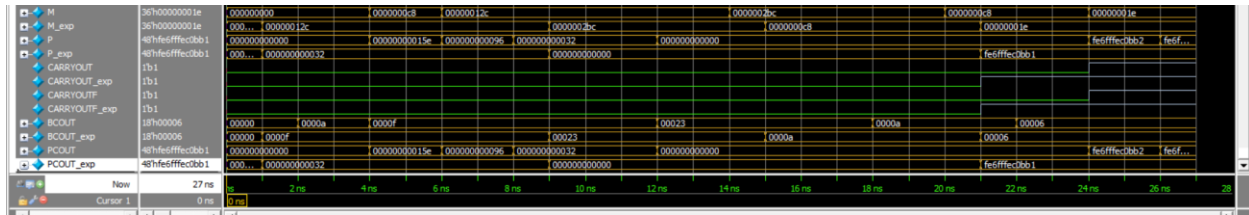
```

vlib work
vlog DSP_REG.v DSP.v DSP_tb.v
vsim -voptargs=+acc work.DSP_tb
add wave *
run -all
#quit -sim

```

QuestaSim Snippets





Synthesis Tool

Constraint File

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top
level signal names in the project

## Clock signal
set_property -dict { PACKAGE_PIN W5    IOSTANDARD LVCMOS33 } [get_ports clk]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]

## Switches
#set_property -dict { PACKAGE_PIN V17    IOSTANDARD LVCMOS33 } [get_ports {sw[0]}]
#set_property -dict { PACKAGE_PIN V16    IOSTANDARD LVCMOS33 } [get_ports {sw[1]}]
#set_property -dict { PACKAGE_PIN W16    IOSTANDARD LVCMOS33 } [get_ports {sw[2]}]
#set_property -dict { PACKAGE_PIN W17    IOSTANDARD LVCMOS33 } [get_ports {sw[3]}]
#set_property -dict { PACKAGE_PIN W15    IOSTANDARD LVCMOS33 } [get_ports {sw[4]}]
#set_property -dict { PACKAGE_PIN V15    IOSTANDARD LVCMOS33 } [get_ports {sw[5]}]
#set_property -dict { PACKAGE_PIN W14    IOSTANDARD LVCMOS33 } [get_ports {sw[6]}]
#set_property -dict { PACKAGE_PIN W13    IOSTANDARD LVCMOS33 } [get_ports {sw[7]}]
#set_property -dict { PACKAGE_PIN V2     IOSTANDARD LVCMOS33 } [get_ports {sw[8]}]
#set_property -dict { PACKAGE_PIN T3     IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2     IOSTANDARD LVCMOS33 } [get_ports
{sw[10]}]
#set_property -dict { PACKAGE_PIN R3     IOSTANDARD LVCMOS33 } [get_ports
{sw[11]}]
#set_property -dict { PACKAGE_PIN W2     IOSTANDARD LVCMOS33 } [get_ports
{sw[12]}]
#set_property -dict { PACKAGE_PIN U1     IOSTANDARD LVCMOS33 } [get_ports
{sw[13]}]
#set_property -dict { PACKAGE_PIN T1     IOSTANDARD LVCMOS33 } [get_ports
{sw[14]}]
#set_property -dict { PACKAGE_PIN R2     IOSTANDARD LVCMOS33 } [get_ports
{sw[15]}]
```

```

## LEDs
#set_property -dict { PACKAGE_PIN U16      IOSTANDARD LVCMOS33 } [get_ports
{led[0]]}
#set_property -dict { PACKAGE_PIN E19      IOSTANDARD LVCMOS33 } [get_ports
{led[1]]}
#set_property -dict { PACKAGE_PIN U19      IOSTANDARD LVCMOS33 } [get_ports
{led[2]]}
#set_property -dict { PACKAGE_PIN V19      IOSTANDARD LVCMOS33 } [get_ports
{led[3]]}
#set_property -dict { PACKAGE_PIN W18      IOSTANDARD LVCMOS33 } [get_ports
{led[4]]}
#set_property -dict { PACKAGE_PIN U15      IOSTANDARD LVCMOS33 } [get_ports
{led[5]]}
#set_property -dict { PACKAGE_PIN U14      IOSTANDARD LVCMOS33 } [get_ports
{led[6]]}
#set_property -dict { PACKAGE_PIN V14      IOSTANDARD LVCMOS33 } [get_ports
{led[7]]}
#set_property -dict { PACKAGE_PIN V13      IOSTANDARD LVCMOS33 } [get_ports
{led[8]]}
#set_property -dict { PACKAGE_PIN V3       IOSTANDARD LVCMOS33 } [get_ports
{led[9]]}
#set_property -dict { PACKAGE_PIN W3       IOSTANDARD LVCMOS33 } [get_ports
{led[10]]}
#set_property -dict { PACKAGE_PIN U3       IOSTANDARD LVCMOS33 } [get_ports
{led[11]]}
#set_property -dict { PACKAGE_PIN P3       IOSTANDARD LVCMOS33 } [get_ports
{led[12]]}
#set_property -dict { PACKAGE_PIN N3       IOSTANDARD LVCMOS33 } [get_ports
{led[13]]}
#set_property -dict { PACKAGE_PIN P1       IOSTANDARD LVCMOS33 } [get_ports
{led[14]]}
#set_property -dict { PACKAGE_PIN L1       IOSTANDARD LVCMOS33 } [get_ports
{led[15]]}

##7 Segment Display
#set_property -dict { PACKAGE_PIN W7      IOSTANDARD LVCMOS33 } [get_ports {seg[0]]}
#set_property -dict { PACKAGE_PIN W6      IOSTANDARD LVCMOS33 } [get_ports {seg[1]]}
#set_property -dict { PACKAGE_PIN U8      IOSTANDARD LVCMOS33 } [get_ports {seg[2]]}
#set_property -dict { PACKAGE_PIN V8      IOSTANDARD LVCMOS33 } [get_ports {seg[3]]}
#set_property -dict { PACKAGE_PIN U5      IOSTANDARD LVCMOS33 } [get_ports {seg[4]]}
#set_property -dict { PACKAGE_PIN V5      IOSTANDARD LVCMOS33 } [get_ports {seg[5]]}
#set_property -dict { PACKAGE_PIN U7      IOSTANDARD LVCMOS33 } [get_ports {seg[6]]}

#set_property -dict { PACKAGE_PIN V7      IOSTANDARD LVCMOS33 } [get_ports dp]

```

```
#set_property -dict { PACKAGE_PIN U2      IOSTANDARD LVCMOS33 } [get_ports {an[0]}]
#set_property -dict { PACKAGE_PIN U4      IOSTANDARD LVCMOS33 } [get_ports {an[1]}]
#set_property -dict { PACKAGE_PIN V4      IOSTANDARD LVCMOS33 } [get_ports {an[2]}]
#set_property -dict { PACKAGE_PIN W4      IOSTANDARD LVCMOS33 } [get_ports {an[3]}]
```

##Buttons

```
set_property -dict { PACKAGE_PIN U18      IOSTANDARD LVCMOS33 } [get_ports rst]
#set_property -dict { PACKAGE_PIN T18      IOSTANDARD LVCMOS33 } [get_ports btnU]
#set_property -dict { PACKAGE_PIN W19      IOSTANDARD LVCMOS33 } [get_ports btnL]
#set_property -dict { PACKAGE_PIN T17      IOSTANDARD LVCMOS33 } [get_ports btnR]
#set_property -dict { PACKAGE_PIN U17      IOSTANDARD LVCMOS33 } [get_ports btnD]
```

##Pmod Header JA

```
#set_property -dict { PACKAGE_PIN J1      IOSTANDARD LVCMOS33 } [get_ports {JA[0]}];#Sch name = JA1
#set_property -dict { PACKAGE_PIN L2      IOSTANDARD LVCMOS33 } [get_ports {JA[1]}];#Sch name = JA2
#set_property -dict { PACKAGE_PIN J2      IOSTANDARD LVCMOS33 } [get_ports {JA[2]}];#Sch name = JA3
#set_property -dict { PACKAGE_PIN G2      IOSTANDARD LVCMOS33 } [get_ports {JA[3]}];#Sch name = JA4
#set_property -dict { PACKAGE_PIN H1      IOSTANDARD LVCMOS33 } [get_ports {JA[4]}];#Sch name = JA7
#set_property -dict { PACKAGE_PIN K2      IOSTANDARD LVCMOS33 } [get_ports {JA[5]}];#Sch name = JA8
#set_property -dict { PACKAGE_PIN H2      IOSTANDARD LVCMOS33 } [get_ports {JA[6]}];#Sch name = JA9
#set_property -dict { PACKAGE_PIN G3      IOSTANDARD LVCMOS33 } [get_ports {JA[7]}];#Sch name = JA10
```

##Pmod Header JB

```
#set_property -dict { PACKAGE_PIN A14      IOSTANDARD LVCMOS33 } [get_ports {JB[0]}];#Sch name = JB1
#set_property -dict { PACKAGE_PIN A16      IOSTANDARD LVCMOS33 } [get_ports {JB[1]}];#Sch name = JB2
#set_property -dict { PACKAGE_PIN B15      IOSTANDARD LVCMOS33 } [get_ports {JB[2]}];#Sch name = JB3
#set_property -dict { PACKAGE_PIN B16      IOSTANDARD LVCMOS33 } [get_ports {JB[3]}];#Sch name = JB4
#set_property -dict { PACKAGE_PIN A15      IOSTANDARD LVCMOS33 } [get_ports {JB[4]}];#Sch name = JB7
```

```

#set_property -dict { PACKAGE_PIN A17      IOSTANDARD LVCMOS33 } [get_ports
{JB[5]}};#Sch name = JB8
#set_property -dict { PACKAGE_PIN C15      IOSTANDARD LVCMOS33 } [get_ports
{JB[6]}};#Sch name = JB9
#set_property -dict { PACKAGE_PIN C16      IOSTANDARD LVCMOS33 } [get_ports
{JB[7]}};#Sch name = JB10

##Pmod Header JC
#set_property -dict { PACKAGE_PIN K17      IOSTANDARD LVCMOS33 } [get_ports
{JC[0]}};#Sch name = JC1
#set_property -dict { PACKAGE_PIN M18      IOSTANDARD LVCMOS33 } [get_ports
{JC[1]}};#Sch name = JC2
#set_property -dict { PACKAGE_PIN N17      IOSTANDARD LVCMOS33 } [get_ports
{JC[2]}};#Sch name = JC3
#set_property -dict { PACKAGE_PIN P18      IOSTANDARD LVCMOS33 } [get_ports
{JC[3]}};#Sch name = JC4
#set_property -dict { PACKAGE_PIN L17      IOSTANDARD LVCMOS33 } [get_ports
{JC[4]}};#Sch name = JC7
#set_property -dict { PACKAGE_PIN M19      IOSTANDARD LVCMOS33 } [get_ports
{JC[5]}};#Sch name = JC8
#set_property -dict { PACKAGE_PIN P17      IOSTANDARD LVCMOS33 } [get_ports
{JC[6]}};#Sch name = JC9
#set_property -dict { PACKAGE_PIN R18      IOSTANDARD LVCMOS33 } [get_ports
{JC[7]}};#Sch name = JC10

##Pmod Header JXADC
#set_property -dict { PACKAGE_PIN J3       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[0]}};#Sch name = XA1_P
#set_property -dict { PACKAGE_PIN L3       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[1]}};#Sch name = XA2_P
#set_property -dict { PACKAGE_PIN M2       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[2]}};#Sch name = XA3_P
#set_property -dict { PACKAGE_PIN N2       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[3]}};#Sch name = XA4_P
#set_property -dict { PACKAGE_PIN K3       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[4]}};#Sch name = XA1_N
#set_property -dict { PACKAGE_PIN M3       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[5]}};#Sch name = XA2_N
#set_property -dict { PACKAGE_PIN M1       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[6]}};#Sch name = XA3_N
#set_property -dict { PACKAGE_PIN N1       IOSTANDARD LVCMOS33 } [get_ports
{JXADC[7]}};#Sch name = XA4_N

##VGA Connector

```

```

#set_property -dict { PACKAGE_PIN G19      IOSTANDARD LVCMOS33 } [get_ports
{vgaRed[0]]}
#set_property -dict { PACKAGE_PIN H19      IOSTANDARD LVCMOS33 } [get_ports
{vgaRed[1]]}
#set_property -dict { PACKAGE_PIN J19      IOSTANDARD LVCMOS33 } [get_ports
{vgaRed[2]]}
#set_property -dict { PACKAGE_PIN N19      IOSTANDARD LVCMOS33 } [get_ports
{vgaRed[3]]}
#set_property -dict { PACKAGE_PIN N18      IOSTANDARD LVCMOS33 } [get_ports
{vgaBlue[0]]}
#set_property -dict { PACKAGE_PIN L18      IOSTANDARD LVCMOS33 } [get_ports
{vgaBlue[1]]}
#set_property -dict { PACKAGE_PIN K18      IOSTANDARD LVCMOS33 } [get_ports
{vgaBlue[2]]}
#set_property -dict { PACKAGE_PIN J18      IOSTANDARD LVCMOS33 } [get_ports
{vgaBlue[3]]}
#set_property -dict { PACKAGE_PIN J17      IOSTANDARD LVCMOS33 } [get_ports
{vgaGreen[0]]}
#set_property -dict { PACKAGE_PIN H17      IOSTANDARD LVCMOS33 } [get_ports
{vgaGreen[1]]}
#set_property -dict { PACKAGE_PIN G17      IOSTANDARD LVCMOS33 } [get_ports
{vgaGreen[2]]}
#set_property -dict { PACKAGE_PIN D17      IOSTANDARD LVCMOS33 } [get_ports
{vgaGreen[3]]}
#set_property -dict { PACKAGE_PIN P19      IOSTANDARD LVCMOS33 } [get_ports Hsync]
#set_property -dict { PACKAGE_PIN R19      IOSTANDARD LVCMOS33 } [get_ports Vsync]

##USB-RS232 Interface
#set_property -dict { PACKAGE_PIN B18      IOSTANDARD LVCMOS33 } [get_ports RsRx]
#set_property -dict { PACKAGE_PIN A18      IOSTANDARD LVCMOS33 } [get_ports RsTx]

##USB HID (PS/2)
#set_property -dict { PACKAGE_PIN C17      IOSTANDARD LVCMOS33    PULLUP true }
[get_ports PS2Clk]
#set_property -dict { PACKAGE_PIN B17      IOSTANDARD LVCMOS33    PULLUP true }
[get_ports PS2Data]

##Quad SPI Flash
##Note that CCLK_0 cannot be placed in 7 series devices. You can access it using
the
##STARTUPE2 primitive.

```

```

#set_property -dict { PACKAGE_PIN D18      IOSTANDARD LVCMOS33 } [get_ports
{QspiDB[0]}]
#set_property -dict { PACKAGE_PIN D19      IOSTANDARD LVCMOS33 } [get_ports
{QspiDB[1]}]
#set_property -dict { PACKAGE_PIN G18      IOSTANDARD LVCMOS33 } [get_ports
{QspiDB[2]}]
#set_property -dict { PACKAGE_PIN F18      IOSTANDARD LVCMOS33 } [get_ports
{QspiDB[3]}]
#set_property -dict { PACKAGE_PIN K19      IOSTANDARD LVCMOS33 } [get_ports QspiCSn]

## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCC0 [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]

create_debug_core u_ila_0 ila
set_property ALL_PROBE_SAME_MU true [get_debug_cores u_ila_0]
set_property ALL_PROBE_SAME_MU_CNT 1 [get_debug_cores u_ila_0]
set_property C_ADV_TRIGGER false [get_debug_cores u_ila_0]
set_property C_DATA_DEPTH 1024 [get_debug_cores u_ila_0]
set_property C_EN_STRG_QUAL false [get_debug_cores u_ila_0]
set_property C_INPUT_PIPE_STAGES 0 [get_debug_cores u_ila_0]
set_property C_TRIGIN_EN false [get_debug_cores u_ila_0]
set_property C_TRIGOUT_EN false [get_debug_cores u_ila_0]
set_property port_width 1 [get_debug_ports u_ila_0/clock]
connect_debug_port u_ila_0/clock [get_nets [list CLK_IBUF_BUFG]]
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe0]
set_property port_width 18 [get_debug_ports u_ila_0/probe0]
connect_debug_port u_ila_0/probe0 [get_nets [list {B_IBUF[0]} {B_IBUF[1]}
{B_IBUF[2]} {B_IBUF[3]} {B_IBUF[4]} {B_IBUF[5]} {B_IBUF[6]} {B_IBUF[7]}
{B_IBUF[8]} {B_IBUF[9]} {B_IBUF[10]} {B_IBUF[11]} {B_IBUF[12]} {B_IBUF[13]}
{B_IBUF[14]} {B_IBUF[15]} {B_IBUF[16]} {B_IBUF[17]}]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe1]
set_property port_width 18 [get_debug_ports u_ila_0/probe1]
connect_debug_port u_ila_0/probe1 [get_nets [list {D_IBUF[0]} {D_IBUF[1]}
{D_IBUF[2]} {D_IBUF[3]} {D_IBUF[4]} {D_IBUF[5]} {D_IBUF[6]} {D_IBUF[7]}
{D_IBUF[8]} {D_IBUF[9]} {D_IBUF[10]} {D_IBUF[11]} {D_IBUF[12]} {D_IBUF[13]}
{D_IBUF[14]} {D_IBUF[15]} {D_IBUF[16]} {D_IBUF[17]}]]
create_debug_port u_ila_0 probe

```



```

set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
set_property port_width 36 [get_debug_ports u_ila_0/probe2]
connect_debug_port u_ila_0/probe2 [get_nets [list {M_OBUF[0]} {M_OBUF[1]}
{M_OBUF[2]} {M_OBUF[3]} {M_OBUF[4]} {M_OBUF[5]} {M_OBUF[6]} {M_OBUF[7]}
{M_OBUF[8]} {M_OBUF[9]} {M_OBUF[10]} {M_OBUF[11]} {M_OBUF[12]} {M_OBUF[13]}
{M_OBUF[14]} {M_OBUF[15]} {M_OBUF[16]} {M_OBUF[17]} {M_OBUF[18]} {M_OBUF[19]}
{M_OBUF[20]} {M_OBUF[21]} {M_OBUF[22]} {M_OBUF[23]} {M_OBUF[24]} {M_OBUF[25]}
{M_OBUF[26]} {M_OBUF[27]} {M_OBUF[28]} {M_OBUF[29]} {M_OBUF[30]} {M_OBUF[31]}
{M_OBUF[32]} {M_OBUF[33]} {M_OBUF[34]} {M_OBUF[35]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
set_property port_width 48 [get_debug_ports u_ila_0/probe3]
connect_debug_port u_ila_0/probe3 [get_nets [list {PCIN_IBUF[0]} {PCIN_IBUF[1]}
{PCIN_IBUF[2]} {PCIN_IBUF[3]} {PCIN_IBUF[4]} {PCIN_IBUF[5]} {PCIN_IBUF[6]}
{PCIN_IBUF[7]} {PCIN_IBUF[8]} {PCIN_IBUF[9]} {PCIN_IBUF[10]} {PCIN_IBUF[11]}
{PCIN_IBUF[12]} {PCIN_IBUF[13]} {PCIN_IBUF[14]} {PCIN_IBUF[15]} {PCIN_IBUF[16]}
{PCIN_IBUF[17]} {PCIN_IBUF[18]} {PCIN_IBUF[19]} {PCIN_IBUF[20]} {PCIN_IBUF[21]}
{PCIN_IBUF[22]} {PCIN_IBUF[23]} {PCIN_IBUF[24]} {PCIN_IBUF[25]} {PCIN_IBUF[26]}
{PCIN_IBUF[27]} {PCIN_IBUF[28]} {PCIN_IBUF[29]} {PCIN_IBUF[30]} {PCIN_IBUF[31]}
{PCIN_IBUF[32]} {PCIN_IBUF[33]} {PCIN_IBUF[34]} {PCIN_IBUF[35]} {PCIN_IBUF[36]}
{PCIN_IBUF[37]} {PCIN_IBUF[38]} {PCIN_IBUF[39]} {PCIN_IBUF[40]} {PCIN_IBUF[41]}
{PCIN_IBUF[42]} {PCIN_IBUF[43]} {PCIN_IBUF[44]} {PCIN_IBUF[45]} {PCIN_IBUF[46]}
{PCIN_IBUF[47]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe4]
set_property port_width 48 [get_debug_ports u_ila_0/probe4]
connect_debug_port u_ila_0/probe4 [get_nets [list {PCOUT_OBUF[0]} {PCOUT_OBUF[1]}
{PCOUT_OBUF[2]} {PCOUT_OBUF[3]} {PCOUT_OBUF[4]} {PCOUT_OBUF[5]} {PCOUT_OBUF[6]}
{PCOUT_OBUF[7]} {PCOUT_OBUF[8]} {PCOUT_OBUF[9]} {PCOUT_OBUF[10]} {PCOUT_OBUF[11]}
{PCOUT_OBUF[12]} {PCOUT_OBUF[13]} {PCOUT_OBUF[14]} {PCOUT_OBUF[15]}
{PCOUT_OBUF[16]} {PCOUT_OBUF[17]} {PCOUT_OBUF[18]} {PCOUT_OBUF[19]}
{PCOUT_OBUF[20]} {PCOUT_OBUF[21]} {PCOUT_OBUF[22]} {PCOUT_OBUF[23]}
{PCOUT_OBUF[24]} {PCOUT_OBUF[25]} {PCOUT_OBUF[26]} {PCOUT_OBUF[27]}
{PCOUT_OBUF[28]} {PCOUT_OBUF[29]} {PCOUT_OBUF[30]} {PCOUT_OBUF[31]}
{PCOUT_OBUF[32]} {PCOUT_OBUF[33]} {PCOUT_OBUF[34]} {PCOUT_OBUF[35]}
{PCOUT_OBUF[36]} {PCOUT_OBUF[37]} {PCOUT_OBUF[38]} {PCOUT_OBUF[39]}
{PCOUT_OBUF[40]} {PCOUT_OBUF[41]} {PCOUT_OBUF[42]} {PCOUT_OBUF[43]}
{PCOUT_OBUF[44]} {PCOUT_OBUF[45]} {PCOUT_OBUF[46]} {PCOUT_OBUF[47]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe5]
set_property port_width 18 [get_debug_ports u_ila_0/probe5]
connect_debug_port u_ila_0/probe5 [get_nets [list {BCOUT_OBUF[0]} {BCOUT_OBUF[1]}
{BCOUT_OBUF[2]} {BCOUT_OBUF[3]} {BCOUT_OBUF[4]} {BCOUT_OBUF[5]} {BCOUT_OBUF[6]}
{BCOUT_OBUF[7]} {BCOUT_OBUF[8]} {BCOUT_OBUF[9]} {BCOUT_OBUF[10]} {BCOUT_OBUF[11]}

```

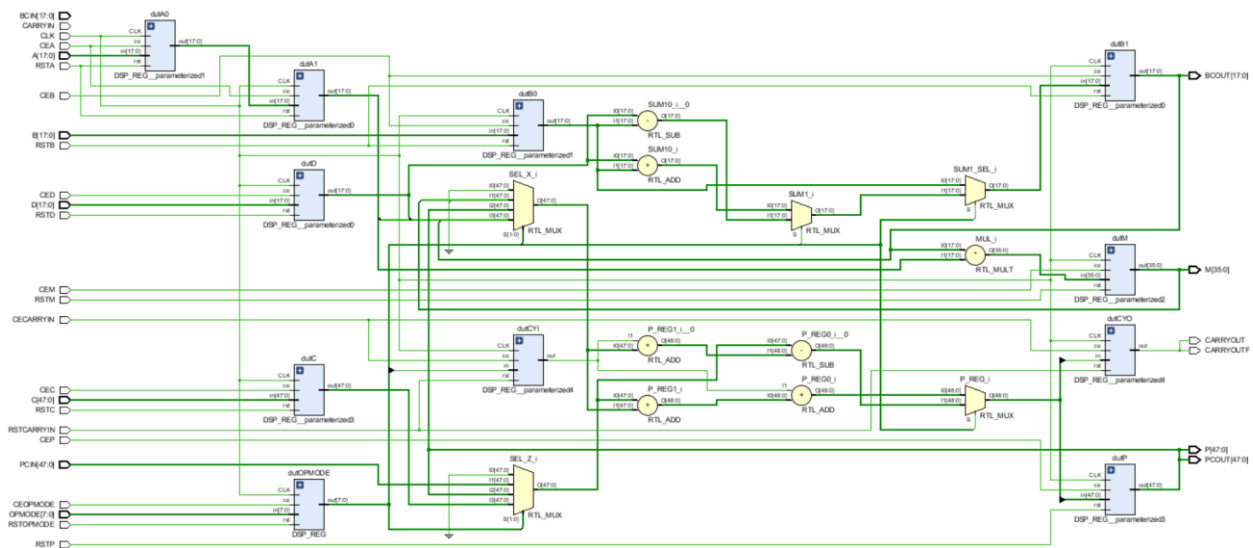
```

{BCOUT_OBUF[12]} {BCOUT_OBUF[13]} {BCOUT_OBUF[14]} {BCOUT_OBUF[15]}
{BCOUT_OBUF[16]} {BCOUT_OBUF[17]]}]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe6]
set_property port_width 8 [get_debug_ports u_ila_0/probe6]
connect_debug_port u_ila_0/probe6 [get_nets [list {OPMODE_IBUF[0]}
{OPMODE_IBUF[1]} {OPMODE_IBUF[2]} {OPMODE_IBUF[3]} {OPMODE_IBUF[4]}
{OPMODE_IBUF[5]} {OPMODE_IBUF[6]} {OPMODE_IBUF[7]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe7]
set_property port_width 18 [get_debug_ports u_ila_0/probe7]
connect_debug_port u_ila_0/probe7 [get_nets [list {A_IBUF[0]} {A_IBUF[1]}
{A_IBUF[2]} {A_IBUF[3]} {A_IBUF[4]} {A_IBUF[5]} {A_IBUF[6]} {A_IBUF[7]}
{A_IBUF[8]} {A_IBUF[9]} {A_IBUF[10]} {A_IBUF[11]} {A_IBUF[12]} {A_IBUF[13]}
{A_IBUF[14]} {A_IBUF[15]} {A_IBUF[16]} {A_IBUF[17]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe8]
set_property port_width 48 [get_debug_ports u_ila_0/probe8]
connect_debug_port u_ila_0/probe8 [get_nets [list {C_IBUF[0]} {C_IBUF[1]}
{C_IBUF[2]} {C_IBUF[3]} {C_IBUF[4]} {C_IBUF[5]} {C_IBUF[6]} {C_IBUF[7]}
{C_IBUF[8]} {C_IBUF[9]} {C_IBUF[10]} {C_IBUF[11]} {C_IBUF[12]} {C_IBUF[13]}
{C_IBUF[14]} {C_IBUF[15]} {C_IBUF[16]} {C_IBUF[17]} {C_IBUF[18]} {C_IBUF[19]}
{C_IBUF[20]} {C_IBUF[21]} {C_IBUF[22]} {C_IBUF[23]} {C_IBUF[24]} {C_IBUF[25]}
{C_IBUF[26]} {C_IBUF[27]} {C_IBUF[28]} {C_IBUF[29]} {C_IBUF[30]} {C_IBUF[31]}
{C_IBUF[32]} {C_IBUF[33]} {C_IBUF[34]} {C_IBUF[35]} {C_IBUF[36]} {C_IBUF[37]}
{C_IBUF[38]} {C_IBUF[39]} {C_IBUF[40]} {C_IBUF[41]} {C_IBUF[42]} {C_IBUF[43]}
{C_IBUF[44]} {C_IBUF[45]} {C_IBUF[46]} {C_IBUF[47]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe9]
set_property port_width 1 [get_debug_ports u_ila_0/probe9]
connect_debug_port u_ila_0/probe9 [get_nets [list CARRYOUTF_OBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe10]
set_property port_width 1 [get_debug_ports u_ila_0/probe10]
connect_debug_port u_ila_0/probe10 [get_nets [list CEA_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe11]
set_property port_width 1 [get_debug_ports u_ila_0/probe11]
connect_debug_port u_ila_0/probe11 [get_nets [list CEB_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe12]
set_property port_width 1 [get_debug_ports u_ila_0/probe12]
connect_debug_port u_ila_0/probe12 [get_nets [list CEC_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe13]

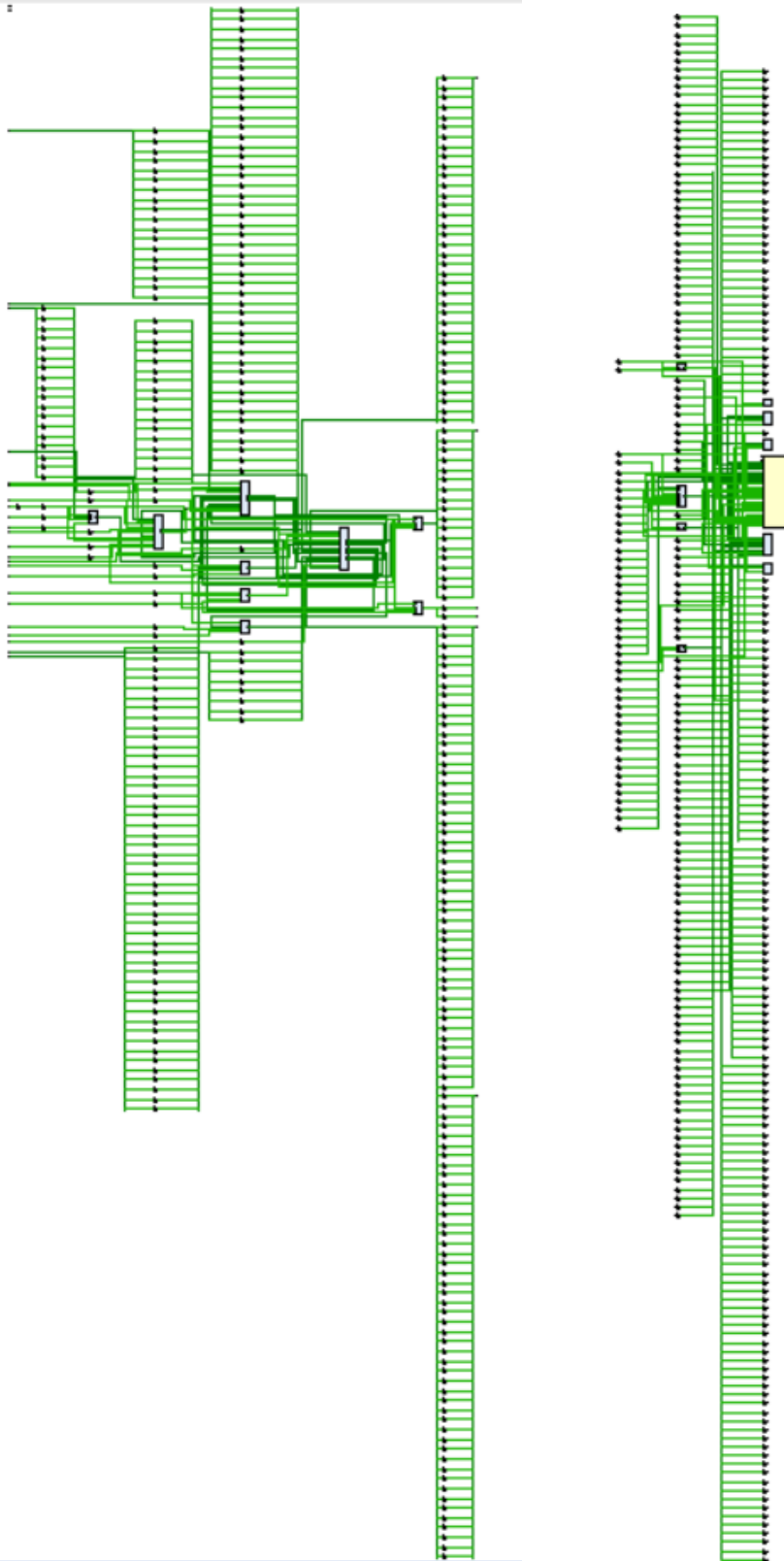
```

```
set_property port_width 1 [get_debug_ports u_ila_0/probe13]
connect_debug_port u_ila_0/probe13 [get_nets [list CECARRYIN_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe14]
set_property port_width 1 [get_debug_ports u_ila_0/probe14]
connect_debug_port u_ila_0/probe14 [get_nets [list CED_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe15]
set_property port_width 1 [get_debug_ports u_ila_0/probe15]
connect_debug_port u_ila_0/probe15 [get_nets [list CEM_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe16]
set_property port_width 1 [get_debug_ports u_ila_0/probe16]
connect_debug_port u_ila_0/probe16 [get_nets [list CEOPMODE_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe17]
set_property port_width 1 [get_debug_ports u_ila_0/probe17]
connect_debug_port u_ila_0/probe17 [get_nets [list CEP_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe18]
set_property port_width 1 [get_debug_ports u_ila_0/probe18]
connect_debug_port u_ila_0/probe18 [get_nets [list CLK_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe19]
set_property port_width 1 [get_debug_ports u_ila_0/probe19]
connect_debug_port u_ila_0/probe19 [get_nets [list RSTA_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe20]
set_property port_width 1 [get_debug_ports u_ila_0/probe20]
connect_debug_port u_ila_0/probe20 [get_nets [list RSTB_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe21]
set_property port_width 1 [get_debug_ports u_ila_0/probe21]
connect_debug_port u_ila_0/probe21 [get_nets [list RSTC_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe22]
set_property port_width 1 [get_debug_ports u_ila_0/probe22]
connect_debug_port u_ila_0/probe22 [get_nets [list RSTCARRYIN_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe23]
set_property port_width 1 [get_debug_ports u_ila_0/probe23]
connect_debug_port u_ila_0/probe23 [get_nets [list RSTD_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe24]
set_property port_width 1 [get_debug_ports u_ila_0/probe24]
```

Elaboration



Synthesis



Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	693	Total Number of Endpoints:	693	Total Number of Endpoints:	NA

There are no user specified timing constraints.

Name	Slice LUTs (134600)	Slice Registers (269200)	DSP s (740)	Bonded IOB (500)	BUFGCTRL (32)
DSP	230	160	1	327	1
dbg_hub (dbg_hub_CV)	0	0	0	0	0
dutA1 (DSP_REG__pa...	0	18	0	0	0
dutB1 (DSP_REG__pa...	0	18	0	0	0
dutC (DSP_REG__par...	0	48	0	0	0
dutCYI (DSP_REG__p...	1	1	0	0	0
dutCYO (DSP_REG__...	0	1	0	0	0
dutD (DSP_REG__par...	16	18	0	0	0
dutM (DSP_REG__par...	1	0	1	0	0
dutOPMODE (DSP_RE...	212	8	0	0	0
dutP (DSP_REG__par...	0	48	0	0	0
u_ila_0 (u_ila_0_CV)	0	0	0	0	0

Tcl Console
Messages
Log
Reports
Design Runs
Utilization
Timing
Debug

☒ Critical warning (6)
☒ Warning (48)
☒ Info (45)
☐ Status (20)
Show All

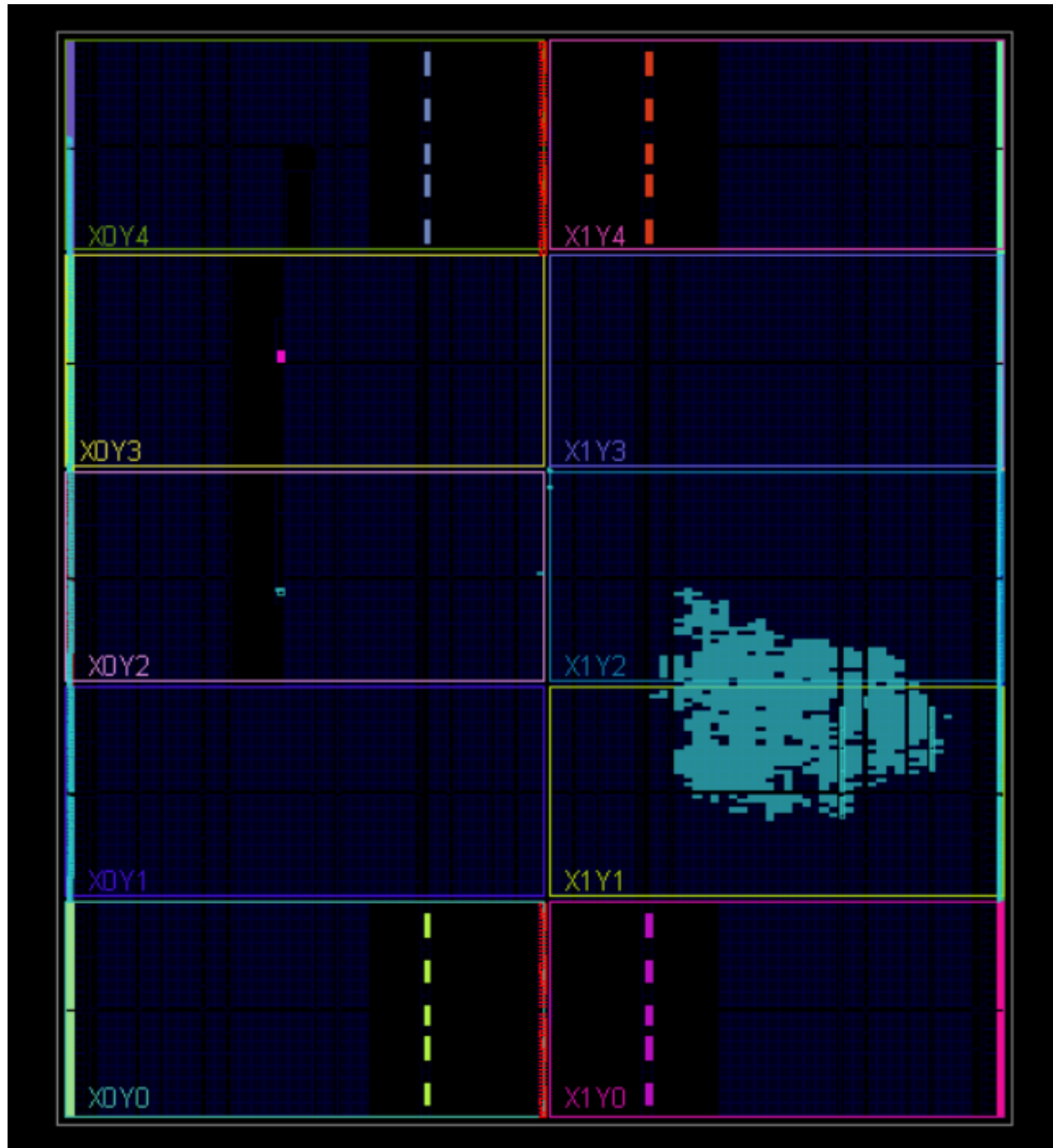
Vivado Commands (3 infos)

General Messages (3 infos)

Synthesis (3 critical warnings, 45 warnings, 33 infos)

Synthesized Design (3 critical warnings, 3 warnings, 9 infos)

Implementation



Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 28.724 ns	Worst Hold Slack (WHS): 0.103 ns	Worst Pulse Width Slack (WPWS): 15.450 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 1036	Total Number of Endpoints: 1028	Total Number of Endpoints: 483

All user specified timing constraints are met.

Name		1	Slice LUTs (133800)	Slice Registers (267600)	F7 Muxes (66900)	F8 Muxes (33450)	Slice (3345 0)	LUT as Logic (133800)	LUT as Memory (46200)	LUT Flip Flop Pairs (133800)	Block RAM Tile (365)	DSP s (740)	Bonded IOB (500)	BUFGCTRL (32)	BSCANEN2 (4)
✓	N	DSP	2664	4225	99	7	1471	2190	474	1563	8	1	327	2	1
>	⚙	dbg_hub (dbg_hub)	474	727	0	0	259	450	24	302	0	0	0	1	1
		[X] dutA1 (DSP_REG___pa...	0	18	0	0	7	0	0	0	0	0	0	0	0
		[X] dutB1 (DSP_REG___pa...	0	18	0	0	5	0	0	0	0	0	0	0	0
		[X] dutC (DSP_REG___par...	0	48	0	0	22	0	0	0	0	0	0	0	0
		[X] dutCY1 (DSP_REG___p...	1	1	0	0	1	1	0	1	0	0	0	0	0
		[X] dutCY0 (DSP_REG___...	0	1	0	0	1	0	0	0	0	0	0	0	0
		[X] dutD (DSP_REG___par...	16	18	0	0	19	16	0	0	0	0	0	0	0
		[X] dutM (DSP_REG___par...	0	0	0	0	0	0	0	0	0	1	0	0	0
		[X] dutOPMODE (DSP_RE...	212	8	0	0	71	212	0	0	0	0	0	0	0
		[X] dutP (DSP_REG___par...	0	48	0	0	12	0	0	0	0	0	0	0	0
>	⚙	u_ila_0 (u_ila_0)	1961	3338	99	7	1147	1511	450	1203	8	0	0	0	0

Questa Lint 2021.1 [L:/mnts/Codes/DSP48A1/questa_lint/int.db]

File Edit View Schematic Window Help

Design

Search: Type Search...

Instance

Module

DSP (11)

DSP

D:_a/Assignments/Codes/DSP48A1/DSP.v

1

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```

module DSP #(
    parameter WIDTH_18 = 18,
    18
    parameter WIDTH_48 = 48,
    48
    parameter A0REG = 0,
    0
    parameter A1REG = 1,
    1
    parameter B0REG = 0,
    0
    parameter B1REG = 1,
    1
    parameter CREG = 1,
    1
    parameter DREG = 1,
    1
    parameter MREG = 1,
    1
    parameter PREG = 1,
    1
    parameter CARRYINREG = 1,
    1
    parameter CARRYOUTREG = 1,
    1

```

Schematic 1

Schematic 1

Lint Summary

Flow Navigator Design DSP_reg.v DSP.v

Lint Checks

Filter: Type here

Waived

Fixed

Pending

2 Uninspected

3 Bug

Verified

Total : 7

Selected : 0

Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Warning	Warning	parameter_name_duplicate		Same parameter name is used in more than one mod... DSP	DSP	Nomenclature...	open	unassign...	1.1 4.3, 3.2 2.2
Warning	Warning	line_char_large		Line has more characters than the specified limit. Curr... none	Rtl Design Style	open	unassign...	3.1 4.5	
Warning	Warning	line_char_large		Line has more characters than the specified limit. Curr... none	Rtl Design Style	open	unassign...	3.1 4.5	
Warning	Warning	line_char_large		Line has more characters than the specified limit. Curr... none	Rtl Design Style	open	unassign...	3.1 4.5	
Warning	Warning	line_char_large		Line has more characters than the specified limit. Curr... none	Rtl Design Style	open	unassign...	3.1 4.5	

Transcript

Message Viewer

Lint Checks

Design Metrics

Design Information

Status History

Lint Dashboard